MSc Computing
Data Mining & Machine Learning

Topics

- Dimensionality Reduction (PCA)
- Clustering: K-means, Hierarchical, DBSCAN

Learning Outcomes

- Reduce feature dimensions
- Apply clustering algorithms effectively

PCA (Principal Component Analysis)

- Reduces high-dimensional data
- Finds principal components
- Removes redundancy
- Improves performance

Clustering Overview

- Unsupervised learning
- Groups similar data points
- No labeled data required

K-means Clustering

- Partitions data into K clusters
- Centroid-based method
- Iterative optimization

Hierarchical Clustering

- Builds tree (dendrogram)
- Agglomerative or Divisive
- No need to predefine clusters

DBSCAN

- Density-based clustering
- Finds arbitrary shape clusters
- Handles noise/outliers well

Week 5 Activities

- Visual clustering demo
- Algorithm comparison
- Discussion on performance

Topics

- Anomaly Detection
- Outlier Analysis
- Neural Networks
- Feedforward & Backpropagation

Learning Outcomes

- Detect anomalies in datasets
- Understand neural network training

Anomaly Detection

- Identify rare or unusual patterns
- Used in fraud detection, security, healthcare

Outlier Analysis

- Detect abnormal data points
- Statistical & ML methods
- Important for data cleaning

Neural Networks

- Inspired by human brain
- Layers: input, hidden, output
- Used in deep learning

Feedforward Process

- Data moves forward through layers
- Weighted sum + activation function
- Produces output

Backpropagation

- Error calculation
- Adjust weights using gradients
- Improves model accuracy

Activities

- Backpropagation explanation
- Simple neural network example